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RESEARCH INTERESTS

Artificial Intelligence and Educational Measurement, Computerized Adaptive Testing, STEM Education

EDUCATION

Beijing Normal University, Beijing, China	Sept. 2024 –Present
Ph.D. Student in Internet Education	
Beijing Normal University, Beijing, China	Sept. 2022 –Aug. 2024
Master Student in Distance Education (transferred to Ph.D. program in Aug. 2024)	
Beijing Normal University, Beijing, China	Sept. 2018 –Jul. 2022
Bachelor of Science in Educational Technology	

PUBLICATIONS

Journal papers (in English)

- Song, Y., Li, H., & Zheng, Q. (2026). Multi-agent automated item generation based on large language models. *Educational Technology & Society*, 29(2), 233–248, [https://doi.org/10.30191/ETS.202604_29\(2\).SP02](https://doi.org/10.30191/ETS.202604_29(2).SP02)
- Du, J., Song, Y., & Zheng, Q. (2026). LLM-simulated nonequivalent groups with anchor test: a novel approach for test equating in the absence of traditional anchor items. *IEEE Transactions on Learning Technologies*, 19, 75–86, <https://doi.org/10.1109/TLT.2026.3661154>
- Song, Y., Du, J., & Zheng, Q. (2025). Automatic item generation for educational assessments: a systematic literature review. *Interactive Learning Environments*, 33(9), 5386–5405, <https://doi.org/10.1080/10494820.2025.2482588>
- Song, Y., Guo, L. & Zheng, Q. (2025). Measuring scientific inquiry ability related to hands-on practice: an automated approach based on multimodal data analysis. *Education and Information Technologies*, 30, 4381–4411, <https://doi.org/10.1007/s10639-024-12991-7>
- Song, Y., Zhu, Q., Wang, H., & Zheng, Q. (2024). Automated essay scoring and revising based on open-source large language models. *IEEE Transactions on Learning Technologies*, 17, 1880–1890, <https://doi.org/10.1109/TLT.2024.3396873>

Journal papers (in Chinese)

- 郑勤华, 宋义深, 李爽. (2026). 中小学生智能素养的表现性评价: 工具设计、数据表征与智能分析. *中国电化教育*, (01), 92–98.
- 杜君磊, 郑勤华, 宋义深. (2025). 大语言模型在无锚题等值中的应用: 以阅读素养测评为例. *武汉大学学报 (理学版)*, 71(05), 591–598. <https://doi.org/10.14188/j.1671-8836.2024.0088>
- 郑勤华, 刘司卓, 宋义深. (2024). 政策视角下综合素质评价的定位与实施路径研究. *中国电化教育*, (12), 81–89.
- 宋义深. (2024). 多模态教育数据在教育实证研究中的应用——基于国际实证研究的系统性文献综述. *开放学习研究*, 29(03), 45–52. <https://doi.org/10.19605/j.cnki.kfxxxyj.2024.03.005>
- 郑勤华, 宋义深. (2024). 互联网教育的发展模式研究——基于利益演化博弈的仿真视角. *中国远程教育*, 44(05), 35–50. <https://doi.org/10.13541/j.cnki.chinade.2024.05.007>
- 宋义深, 冯晓英. (2023). 在线学习失信行为的驱动因素与形成机制研究——基于扎根理论. *电化教育研究*, 44(09), 64–70. <https://doi.org/10.13811/j.cnki.eer.2023.09.009>
- 傅骞, 宋义深, 郑娅峰. (2023). 面向信息科技教育的物联网平台研究——MixIO 物联网平台的设计与实现. *电化教育研究*, 44(05), 99–103+128. <https://doi.org/10.13811/j.cnki.eer.2023.05.013>

Conference papers

- Song, Y. (2025). CoScore: an LLM-empowered human-AI collaborative text scoring rule generation and calculation engine. *IEEE International Conference on Teaching, Assessment, and Learning for Engineering (IEEE TALE)*, <https://doi.org/10.1109/TALE66047.2025.11346772>

PROJECTS

MixIO: An education-first open-source IoT platform

Main Contributor

Beijing, China

Aug. 2020 –Present

- Developed the front-end and back-end of an IoT Server for K-12 IT education.
- Maintained an open source repository: <https://gitee.com/mixly2/mixio><https://gitee.com/mixly2/mixio>.
- The official server has around 1,000 active users per month, supporting students in building their own IoT Apps in a low-code manner: <https://mixio.mixly.cn/><https://mixio.mixly.cn/>.

EXPERIENCE

Intern Information Technology Teacher

Beijing Xicheng Foreign Languages School (BFLS)

Beijing, China

Oct. 2021 –Dec. 2021

- Taught Python programming to first-year high school students.
- Designed teaching activities and produced multimedia teaching materials.

AWARDS

- **National Scholarship for Doctoral Students of China** Dec. 2024
Ministry of Education, China
- **National Scholarship for Undergraduate Students of China** Dec. 2021
Ministry of Education, China

SKILLS

- **IELTS (Academic): 7.5** (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.0)
- Python (proficient), JavaScript (proficient), R (familiar), Stata (familiar)